Homework 6: Thin Films

Constants used. $k = 1.381 \times 10^{-23}$ J/K $= 8.617 \times 10^{-5}$ eV/K; 1 amu $= 1.661 \times 10^{-27}$ kg; $N_A = 6.022 \times 10^{23}$ mol^{-1}; 1 torr = 133.32 Pa.

Vapor pressures. These were read from the vapor-pressure chart attached to the homework (Alcock et al.):

Vapor pressure (torr)	10^{-6}	10^{-5}	10^{-4}	10^{-3}	10^{-2}	3×10^{-2}
Al	829 °C	901 °C	999 °C	1115 °C	1248 °C	1321 °C
Au	958 °C	1041 °C	1151 °C	1281 °C	1429 °C	1511 °C

Between these points, log P is interpolated linearly in 1/T (Clausius–Clapeyron).

1. Crucible temperature for equal Al and water-vapor fluxes (10 pts)

The evaporator in Example 12.2. Al charge at 1100 °C, where $P_e \approx 1 \times 10^{-3}$ torr. Planetary radius $r_0 = 40$ cm. Crucible diameter 5 cm, so A = 19.6 cm^2. Background of 10^{-6} torr water vapor at room temperature. $\rho_{Al} = 2700$ kg/m^3. The example finds R = 2.9×10^{-11} m/s (17.8 Å/min), $J_{Al} = 1.78 \times 10^{18}$ atoms/(m^2·s) and $J_{H_2O} = 4.8 \times 10^{18}$ molecules/(m^2·s). So at 1100 °C, aluminum atoms arrive only 0.37 times as often as water molecules.

Step 1: the water flux, which is the target. M = 18, T = 300 K, P = 10^{-6} torr = 1.333×10^{-4} Pa:

$$J_{H_2O} = P / \sqrt{(2\pi mkT)} = 1.333 \times 10^{-4} / \sqrt{(2\pi \times 18 \times 1.661 \times 10^{-27} \times 1.381 \times 10^{-23} \times 300)} = \mathbf{4.78 \times 10^{18}\ molecules/(m^2 \cdot s)}$$

Step 2: the growth rate this requires. Arrival rate = growth rate × number density ($J_{Al} = R \cdot \rho / m_{Al}$). Setting $J_{Al} = J_{H_2O}$:

$$R = J_{H_2O} \cdot m_{Al} / \rho = (4.78 \times 10^{18})(27 \times 1.661 \times 10^{-27}\ \text{kg}) / 2700\ \text{kg/m}^3 = 7.9 \times 10^{-11}\ \text{m/s}$$

> **Growth rate:** R ≈ **7.9×10^{-11} m/s ≈ 0.8 Å/s ≈ 48 Å/min**. That is about 2.7 times the example's 17.8 Å/min.

Step 3: the vapor pressure that gives this rate (Campbell Eq. 12.8: the evaporated mass spreads evenly over the planetary sphere):

$$R = A \cdot \sqrt{(m / 2\pi kT)} \cdot P_e / (4\pi {r_0}^2 \rho)$$

$$\Longrightarrow\ P_e = 4\pi {r_0}^2 \rho R / [A \sqrt{(m / 2\pi kT)}] = (4\pi \times 0.40^2 \times 2700 \times 7.9 \times 10^{-11}) / [19.6 \times 10^{-4} \times \sqrt{(m / 2\pi kT)}]$$

The numerator is 4.30×10^{-7} kg/s. With T ≈ 1444 K, $\sqrt{(m/2\pi kT)} = 5.98 \times 10^{-4}$ s/m, so

$$P_e = 4.30 \times 10^{-7} / (19.6 \times 10^{-4} \times 5.98 \times 10^{-4}) = 0.367\ \text{Pa} = \mathbf{2.7 \times 10^{-3}\ torr}$$

A quicker way to see the same thing: the rate must go up by a factor 4.78/1.78 = 2.7, so $P_e/\sqrt{T}$ must also rise by 2.7×, from 1×10^{-3} torr to about 2.7×10^{-3} torr (the √T correction is under 3%).

Step 4: read the temperature off the vapor-pressure chart. Al reaches 2.7×10^{-3} torr at about 1170 °C. (The same chart puts 1×10^{-3} torr at about 1115 °C, matching the example's "about 1×10^{-3} torr at 1100 °C".)

> **Answer:** the crucible must be at **T ≈ 1170 °C (≈ 1440 K)**, about 70 °C hotter than in Example 12.2. The Al then grows at **≈ 0.8 Å/s (≈ 48 Å/min)**.

What this means. Even at 1170 °C, about one water molecule strikes the wafer for every Al atom that arrives. If every water molecule stuck, the film could hold oxygen and hydrogen at a level comparable to the aluminum itself. Water does not stick perfectly, but this is why real evaporations use either a much higher deposition rate (a hotter source) or a lower base pressure (10^{-7} torr or below) to get clean, low-resistivity metal. A small temperature increase helps a lot, because vapor pressure rises roughly exponentially with temperature.

2. Au/Al co-evaporation with a rotating shutter (10 pts)

Timing. At 10 rpm one revolution takes 6 s. The opening is half the shutter, so each source deposits for **3 s per cycle**. One 3 Å monolayer per cycle therefore needs a deposition rate, while the source is uncovered, of:

$$R_d = 3\ \text{Å} / 3\ \text{s} = 1\ \text{Å/s} = 1 \times 10^{-10}\ \text{m/s} \quad \text{(for both Au and Al)}$$

Required evaporation rate. On a deposition sphere of radius r_0 the deposition is uniform (Q4 and Campbell Eq. 12.8, the same formula Example 12.2 uses): $R_d = R_{ev} / (4\pi r_0^2 \rho)$. So

$$R_{ev} = 4\pi r_0^2\, \rho\, R_d, \quad \text{with } 4\pi r_0^2 = 4\pi(0.50\ \text{m})^2 = 3.14\ \text{m}^2$$

- Au: $R_{ev} = 3.14 \times 18{,}900 \times 10^{-10}$ = **5.94×10^{-6} kg/s**
- Al: $R_{ev} = 3.14 \times 2{,}700 \times 10^{-10}$ = **8.48×10^{-7} kg/s**

Required vapor pressure (Hertz–Knudsen, A = 10 cm^2 = 10^{-3} m^2):

$$R_{ev} = A \cdot \sqrt{(m / 2\pi kT)} \cdot P_e \implies P_e = R_{ev} / [A \sqrt{(m / 2\pi kT)}]$$

Solve this together with each vapor-pressure curve, as in Q1:

	Atom mass m	Required P_e	Source temperature (from the chart)
Gold	197 amu = 3.27×10^{-25} kg	4.1 Pa = 3.1×10^{-2} torr	**≈ 1510 °C (≈ 1780 K)**
Aluminum	27 amu = 4.48×10^{-26} kg	1.46 Pa = 1.1×10^{-2} torr	**≈ 1250 °C (≈ 1530 K)**

Check for Au: $\sqrt{(m/2\pi kT)} = \sqrt{(3.27 \times 10^{-25} / (2\pi \times 1.381 \times 10^{-23} \times 1782))} = 1.46 \times 10^{-3}$ s/m, so $P_e = 5.94 \times 10^{-6} / (10^{-3} \times 1.46 \times 10^{-3}) = 4.1$ Pa. ✔

> **Answer:** run the **Au source at about 1510 °C** and the **Al source at about 1250 °C**. Each then deposits 1 Å/s while uncovered, which is one 3 Å layer per shutter revolution.

Gold needs a much hotter source for two reasons. Matching the same thickness rate takes 7 times more mass per second (its density is 7 times higher). And gold's vapor pressure is lower than aluminum's at any given temperature. Equal thicknesses also give a nearly 50/50 atomic mixture: Au has 5.78×10^{22} atoms/cm^3 and Al has 6.03×10^{22}, a 49:51 ratio.

3. Reaction-rate-limited deposition: 700 °C → 800 °C (10 pts)

In the reaction-rate-limited regime, the deposition rate follows an Arrhenius law, $R \propto \exp(-E_a/kT)$. So

$$R(800\ °C) / R(700\ °C) = \exp[\ (E_a/k)(1/T_1 - 1/T_2)\ ]$$

$$= \exp[\ (2\ eV / 8.617 \times 10^{-5}\ eV/K)(1/973\ K - 1/1073\ K)\ ] = \exp(23{,}210 \times 9.58 \times 10^{-5}) = \exp(2.22) \approx 9.2$$

Prediction: R(800 °C) ≈ 9.2 × 1000 Å/min ≈ **9,200 Å/min (about 0.9 µm/min)**.

If the measured rate is much lower: the process has most likely moved into the **mass-transport-limited** (gas-phase diffusion-limited) regime. The surface reaction has become so fast that the rate is now set by how quickly reactant gas can diffuse through the boundary layer to the wafer. Diffusion depends only weakly on temperature (roughly $T^{1.5–2}$), so the rate no longer increases exponentially. Other possibilities are depletion of the reactant along the reactor, or gas-phase reactions (homogeneous nucleation) using up reactant before it reaches the wafer.

How to prove it:

1. **Arrhenius plot.** Measure the rate at several temperatures between 600 and 900 °C and plot ln(rate) against 1/T. A reaction-limited process gives a straight line with slope $-E_a/k$ (−2 eV). If the slope flattens above some temperature, the process has become transport-limited there.
2. **Change the gas flow and pressure at 800 °C.** A transport-limited rate increases with gas velocity, since a thinner boundary layer ($\delta \propto 1/\sqrt{v}$) lets reactant reach the wafer faster. It also changes with total pressure and with the reactant's partial pressure. A reaction-limited rate does not depend on flow.
3. **Check uniformity.** In the transport-limited regime, the rate depends on wafer position and spacing. Wafers further downstream, or more tightly packed, grow thinner films. In the reaction-limited regime, all wafers at the same temperature grow at the same rate.

4. Uniform deposition on a sphere (10 pts)

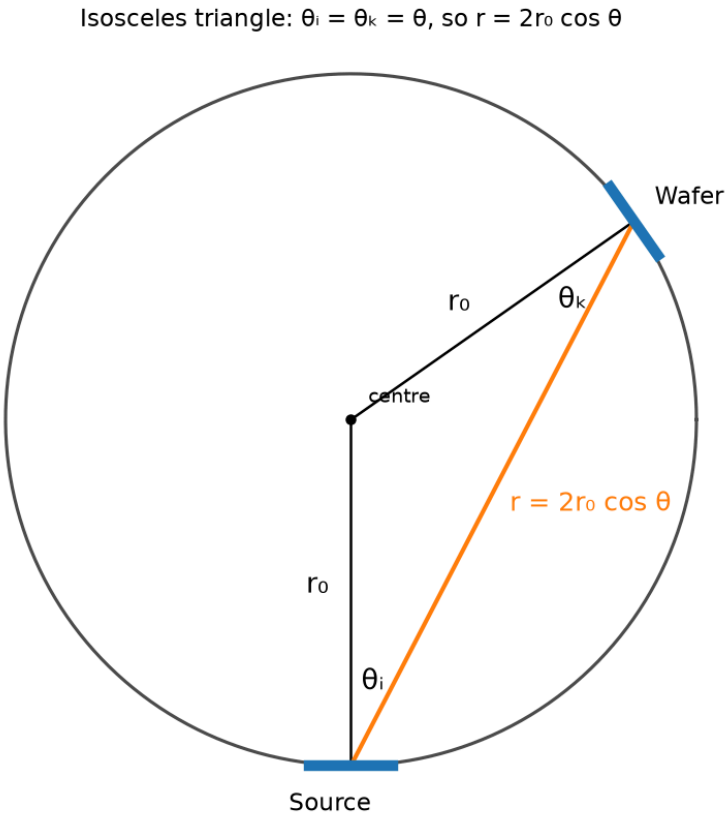


Figure 1. A small planar source and a wafer, both on the inside of a sphere of radius r_0 and both facing its centre.

Emission from a small planar source (Knudsen cosine law). The source loses mass at a total rate R_{ev}. The mass sent into a small solid angle dω at angle θ_i from the source's normal is:

$$dR = (R_{ev} / \pi) \cos \theta_i \, d\omega$$

(The 1/π normalizes it: ∫ cos θ dω over the half-space in front of the source equals π.)

Mass arriving at the wafer. A wafer area element dA_k at distance r, whose normal makes angle θ_k with the line to the source, covers a solid angle $d\omega = \cos \theta_k \, dA_k / r^2$. So the mass deposition rate per unit wafer area is:

$$dR / dA_k = (R_{ev} / \pi) \cdot \cos \theta_i \cos \theta_k / r^2$$

Geometry on the sphere. The source and the wafer both lie on the sphere and face its centre, so each normal points along a radius. The triangle formed by the source, the wafer and the centre has two sides equal to r_0, so it is isosceles. Its base angles are equal, so $\boldsymbol{\theta_i = \theta_k = \theta}$. Dropping a perpendicular from the centre to the chord splits the chord in half: $r/2 = r_0 \cos \theta$, so

$$r = 2r_0 \cos \theta$$

Substitute:

$$dR / dA_k = (R_{ev} / \pi) \cdot \cos^2\theta / (4r_0^2 \cos^2\theta) = R_{ev} / (4\pi r_0^2)$$

The $\cos^2\theta$ terms cancel, so the result **does not depend on θ**, the wafer's position on the sphere. The thickness growth rate is

$$R_d = R_{ev} / (4\pi r_0^2 \rho)$$

This is the same for every wafer on the sphere. ∎

Physically, wafers far from the source are farther away (r is larger), but they face the source more squarely and the source sends more material toward them, and these effects exactly balance. This is why planetary evaporators mount wafers on a spherical dome that passes through the source. The result assumes a small planar source with an ideal cosine emission pattern. Real e-beam and boat sources are somewhat more directional, so planetary rotation is also used.

Sources

- S. A. Campbell, *Fabrication Engineering at the Micro- and Nanoscale*, Ch. 12 (Physical Deposition: Evaporation and Sputtering), Eq. 12.2 and the deposition-rate equations.
- Vapor-pressure chart attached to the homework (data adapted from Alcock et al.). The values were read off the Al and Au curves.
- Kurt J. Lesker Company, aluminum evaporation notes (cross-check: Al at 10^{-4} torr ≈ 1010 °C): https://www.lesker.com/newweb/deposition_materials/deposition-materials-notes.cfm?pgid=al1